

YOUSSEF ASHMAWY

054-324-1340 | ashmawyyoussif@gmail.com | linkedin.com/in/youssif-ashmawy | github.com/Youssif-Ashmawy | youssif-ashmawy.github.io/Portfolio

SCE-ID-1207094

Education

Carleton University

Bachelor of Computer Science Honours with High Distinction

Sep 2022 – Aug 2025

Ottawa, Canada

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL, HTML, CSS, Scheme

Cloud/DevOps: AWS, Nutanix, Azure, Docker, Kubernetes, Linux, CI/CD, GitHub Actions, Velero, OpenStack

Frameworks: React, Node.js, PyTest, FastAPI, JWT authentication

Databases/Object Storage: PostgreSQL, SQLite, MongoDB, ChromaDB, MinIO, S3 Bucket

AI/ML: PyTorch, Transformers, Diffusers, Scikit-Learn, OpenCV, RAG, Claude, MCP, MLflow

Developer Tools: Git, Jupyter Notebook, Postman, Grafana, NetworkX, NetGraph, Seaborn, Kafka

Experience

DataScience ME

MLOps Engineer

Jun 2026 – Present

Riyadh, Saudi Arabia

- Optimized latency of locally-hosted LLMs by fine-tuning inference parameters using **vLLM**, reduced by **90%**.
- Built a **Grafana** dashboard leveraging **vLLM** and **DCGM** metrics to monitor LLM performance and GPU utilization.
- Redesigning **GPU allocation** and request **load balancing** to scale concurrent-user throughput.
- Developing a **CI/CD**-style pipeline in **Dataiku** to manage model/app deployment.

Kuberox Technologies

AI Systems Engineer

Sep 2025 – May 2026

Riyadh, Saudi Arabia

- Deployed and configured **Nutanix** hyperconverged **Kubernetes** and **AI** platforms, accelerating production readiness.
- Designed two-phase **Velero** migration workflow, reducing downtime by **50%**.
- Prompt engineered** staged descriptive queries for **Claude**, optimizing responses and reducing token usage.

Hydro Ottawa

Automation Engineer

May 2024 – Aug 2025

Ottawa, Canada

- Automated GSD reports (**Python**, **Node.js**, **AWS**), cutting manual effort by **75%** and improving accuracy by **30%**.
- Built on-demand dashboards and reporting tools, reducing report latency by **60%**.
- Developed automated retention workflows via **Iron Mountain API**, improving compliance readiness by **40%**.

Projects

TactileNet Modified

Python, PyTorch, Transformers, Diffusers

GitHub Repo | Research Paper

- Expanded tactile dataset and integrated multimodal feedback using **Transformers**, boosting model robustness by **25%**.
- Optimized training pipeline with **PyTorch**, **PEFT**, **LoRA**, and **Bitsandbytes**, improving generalization.
- Benchmarked multiple **Hugging Face** models to select the optimal architecture, improving inference quality by **20%**.

RAG ME

Python, LangChain, Groq API, RAG, Streamlit, fastembed, Hybrid Search, Docker

GitHub Repo | Live Demo

- Built GitHub RAG chatbot using **LangChain**, **Groq (Llama 3.3 70B)**, and **fastembed** for GPU-free code Q&A.
- Implemented hybrid retrieval with **BM25** + **cosine similarity** and **Reciprocal Rank Fusion** for higher accuracy.
- Deployed via **Docker** + **Streamlit** with local **ONNX embeddings** for zero-cost hosting.

MLOps Bias Monitoring Pipeline

Python, Kafka, MLflow, Docker, Optuna, XGBoost, Fairlearn, Streamlit

GitHub Repo | Project Report

- Designed real-time ML system with **Kafka** + **XGBoost** in producer-consumer architecture for scalable inference.
- Applied **Fairlearn** for bias monitoring, reducing Equalized Odds Difference **1.0** → **0.10** while maintaining **85%** accuracy.
- Built MLOps pipeline with **MLflow** tracking, **Optuna** (30 trials), and **Streamlit** dashboard for bias monitoring.

Certifications

Microsoft Certified: Azure AI Fundamentals

Nutanix Cloud Certified: NCA, NCP-MCI, NCP-CN, NCP-MCA, NCP-AI

Red Hat Certified System Administrator (RHCSA)

Google Cybersecurity Professional Certificate